

Guan-Horng Liu

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RESEARCH INTERESTS

Generative AI, AI for science, Boltzmann distribution sampling, Schrödinger bridge, dynamic optimal transport, stochastic optimal control, forward-backward stochastic differential equation, matching-based method

RESEARCH EXPERIENCE

FAIR, Meta AI

Research Scientist

Research Scientist Intern

New York, NY

09/2024 - present

Summer 2023

NVIDIA Research

Research Intern

Santa Clara, CA

Summer 2022

Uber Advanced Technology Group

Robotics Research Engineer

Pittsburgh, PA

09/2017 – 12/2018

Aptiv Mobility Group

Research Intern

Pittsburgh, PA

Summer 2016

EDUCATION

Georgia Institute of Technology

Ph.D. in Machine Learning (GPA: 4.0/4.0)

◦ Advisor: Evangelos A. Theodorou

◦ Received ICLR'21 Spotlight, ICML'21 Oral, NeurIPS'21 Spotlight, and NeurIPS'22 Oral

Atlanta, GA

07/2024

Carnegie Mellon University

M.S. in Robotics (GPA: 4.0/4.0)

◦ Advisor: George Kantor

Pittsburgh, PA

05/2017

Tokyo Institute of Technology

Research Exchange Program (GPA: 4.0/4.0)

◦ Advisor: Edwardo F. Fukushima

Tokyo, Japan

06/2014

National Taiwan University

B.S. in Mechanical Engineering (GPA: 3.99/4.0)

◦ Advisor: Pei-Chun Lin

◦ Graduated Cum Laude; Best Paper Award in 2013 IEEE/SICE ISS

Taipei, Taiwan

06/2013

Last update: 09/2025

PUBLICATIONS

(*Equal contribution *Core contributors †Equal advising ‡Alphabetical order)

Conference Papers

- [C1] Adjoint Schrödinger Bridge Sampler **[Oral, 0.3%]**
G.-H. Liu*, J. Choi*, Y. Chen, B. K. Miller, Ricky T. Q. Chen*
Advances in Neural Information Processing Systems (NeurIPS), 2025
- [C2] Non-equilibrium Annealed Adjoint Sampler
J. Choi*, Y. Chen, M. Tao, **G.-H. Liu***
Advances in Neural Information Processing Systems (NeurIPS), 2025
- [C3] Momentum Multi-Marginal Schrödinger Bridge Matching
P. Theodoropoulos, A. D. Saravanos, E. Theodorou†, **G.-H. Liu†**
Advances in Neural Information Processing Systems (NeurIPS), 2025
- [C4] MDNS: Masked Diffusion Neural Sampler via Stochastic Optimal Control
Y. Zhu*, W. Guo*, J. Choi, **G.-H. Liu**, Y. Chen, M. Tao
Advances in Neural Information Processing Systems (NeurIPS), 2025
- [C5] Adjoint Sampling: Highly-Scalable Diffusion Samplers via Adjoint Matching
A. J. Havens*, B. K. Miller*, B. Yan*, C. Domingo-Enrich, A. Sriram, D. S. Levine, B. M. Wood, B. Hu, B. Amos, B. Karrer, X. Fu*, **G.-H. Liu***, Ricky T. Q. Chen*
International Conference on Machine Learning (ICML), 2025
- [C6] Feedback Schrödinger Bridge Matching **[Oral, 1.8%]**
P. Theodoropoulos, N. Komianos, V. Pacelli, **G.-H. Liu†**, E. Theodorou†
International Conference on Learning Representations (ICLR), 2025
- [C7] Generalized Schrödinger Bridge Matching
G.-H. Liu, Y. Lipman, M. Nickel, B. Karrer, E. Theodorou, Ricky T. Q. Chen
International Conference on Learning Representations (ICLR), 2024
- [C8] A Robust Differential Neural ODE Optimizer
P. Theodoropoulos, **G.-H. Liu**, T. Chen, A. D. Saravanos, E. Theodorou
International Conference on Learning Representations (ICLR), 2024
- [C9] Mirror Diffusion Models for Constrained and Watermarked Generation
G.-H. Liu, T. Chen, E. Theodorou†, M. Tao†
Advances in Neural Information Processing Systems (NeurIPS), 2023
- [C10] Deep Momentum Multi-Marginal Schrödinger Bridge
T. Chen, **G.-H. Liu**, M. Tao, E. Theodorou
Advances in Neural Information Processing Systems (NeurIPS), 2023
- [C11] I²SB: Image-to-Image Schrödinger Bridge
G.-H. Liu, A. Vahdat, D.-A. Huang, E. Theodorou, W. Nie†, A. Anandkumar†
International Conference on Machine Learning (ICML), 2023
- [C12] Deep Generalized Schrödinger Bridge **[Oral, 1.9%]**
G.-H. Liu, T. Chen*, O. So*, E. Theodorou
Advances in Neural Information Processing Systems (NeurIPS), 2022
- [C13] Likelihood Training of Schrödinger Bridge using Forward-Backward SDEs Theory
T. Chen*, **G.-H. Liu***, E. Theodorou
International Conference on Learning Representations (ICLR), 2022

- [C14] Second-Order Neural ODE Optimizer [Spotlight, 3.0%]
G.-H. Liu, T. Chen, E. Theodorou
Advances in Neural Information Processing Systems (NeurIPS), 2021
- [C15] Dynamic Game Theoretic Neural Optimizer [Long talk, 3.0%]
G.-H. Liu, T. Chen, E. Theodorou
International Conference on Machine Learning (ICML), 2021
- [C16] Differential Dynamic Programming Neural Optimizer [Spotlight, 3.8%]
G.-H. Liu, T. Chen, E. Theodorou
International Conference on Learning Representations (ICLR), 2021
- [C17] Variational Inference MPC using Tsallis Divergence
Z. Wang*, O. So*, J. Gibson, B. Vlahov, M. S. Gandhi, **G.-H. Liu**, E. Theodorou
Robotics: Science and Systems (RSS), 2021
- [C18] Learning End-to-end Multimodal Sensor Policies for Autonomous Navigation
G.-H. Liu, A. Siravuru, S. Prabhakar, M. Veloso, G. Kantor
Conference on Robot Learning (CoRL), 2017
- [C19] Autonomous Control of the WAM-V Catamaran Type USV: Propulsion System Design
G.-H. Liu, A. Y. Yasutomi, A. Holgado, E. F. Fukushima
Annual Conference of the Robotics Society of Japan, 2014
- [C20] Design of a Kangaroo Robot with Dynamic Jogging Locomotion [Best Paper Award, 0.8%]
G.-H. Liu, H.-Y. Lin, H.-Y. Lin, S.-T. Chen, P.-C. Lin
IEEE/SICE International Symposium on System Integration (ISS), 2013

Journal Papers

- [J1] Optimal Transport for Generating Transition States in Chemical Reactions
C. Duan*, **G.-H. Liu***, Y. Du*, T. Chen, Q. Zhao, H. Jia, C. P. Gomes, E. Theodorou, H. J. Kulik
Nature Machine Intelligence, 2025
- [J2] Improving Generative Model-based Unfolding with Schrödinger Bridges
S. Diefenbacher[‡], **G.-H. Liu**[‡], V. Mikuni[‡], B. Nachman[‡], W. Nie[‡]
Physical Review D, 2024
- [J3] A Bio-Inspired Hopping Kangaroo Robot with an Active Tail
G.-H. Liu, H.-Y. Lin, H.-Y. Lin, S.-T. Chen, P.-C. Lin
Journal of Bionic Engineering, 2014

Workshop Papers & Technical Reports

- [W1] Large-Scale Optimization for DNN Architecture: A Dynamical System Theory
G.-H. Liu
Georgia Tech Machine Learning Ph.D. Thesis, 2024
- [W2] Augmented Bridge Matching
V. D. Bortoli, **G.-H. Liu**, T. Chen, E. Theodorou, W. Nie
Preprint, 2023
- [W3] Improved Sampling via Learned Diffusions
L. Richter*, J. Berner*, **G.-H. Liu**
ICML Workshop on New Frontiers in Learning, Control, Dynamical Systems, 2023

- [W4] Spatio-Temporal Differential Dynamic Programming for Control of Fields
E. N. Evans, O. So, A. P Kendall, **G.-H. Liu**, E. Theodorou
Preprint, 2021
- [W5] Deep Learning Theory Review: An Optimal Control and Dynamical Systems Perspective
G.-H. Liu, E. Theodorou
Preprint, 2019
- [W6] High Dimensional Planning and Learning for Off-Road Driving
G.-H. Liu
CMU Robotics Institute Master Thesis, 2017

PATENTS

Conditional Diffusion Model for Data-to-Data Translation

U.S. Patent Application No. 18/431,527 08/2024

HONORS & AWARDS

Fellowships & Scholarships

AE Graduate Research Fellowship, Georgia Tech 2022 – 2023
Study Abroad Scholarship, Ministry of Education, Taiwan 2019 – 2021
Student Exchange Scholarship, JASSO, Japan 2013

Awards & Prizes

Best Paper Award, IEEE/SICE ISS 2013
Third Prize, Chuian-Yan Thesis Paper Competition, Taiwan 2013
Presidential Awards ($\times 4$), Top 5% in National Taiwan University 2009 – 2014

PRESS COVERAGE

Meta AI Blog: A breakthrough in highly scalable, reward-driven generative modeling 05/2025
Chemistry World: AI predicts transition states with exceptional precision 05/2025
MIT News: New model predicts a chemical reaction’s point of no return 04/2025
Nature Machine Intelligence: Cover photo on Volume 7 Issue 4 04/2025

INVITED TALKS

Sampling with Schrödinger Bridge: An Adjoint-Matching Perspective

NeurIPS (Oral talk, 0.3%) 12/2025
Flatiron Institute, Center for Computational Mathematics 08/2025
Microsoft Research, Generative Modeling & Sampling Seminar 07/2025

Learning Scalable Diffusion Models using Optimality and Constraint Structures

Learning on Graphs NYC Meetup 11/2024
CS5785 Applied Machine Learning, Cornell Tech (Host: Brandon Amos) 09/2024
AMLab, University of Amsterdam 05/2024

Sony AI	04/2024
National Taiwan University (Host: Shao-Hua Sun)	04/2024
Appier Group Inc.	01/2024
School of Industrial and Systems Engineering, Georgia Tech (Host: Yao Xie)	12/2023
FAIR, Meta AI	11/2023
Nvidia Research	10/2023

Mirror Diffusion Models

Learning on Graphs and Geometry Reading Group	10/2023
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(Generalized) Schrödinger Bridge

AE8803 Optimal Transport Theory & Applications, Georgia Tech (Host: Yongxin Chen) ...	04/2025
Learning on Graphs and Geometry Reading Group	02/2023
NeurIPS Workshop on Score-Based Methods	12/2022
Rough Path Interest Group, Alan Turing Institute	11/2022
IBM Research Seminar	11/2022
School of Mathematics, Georgia Tech (Host: Molei Tao)	11/2021

Optimal Control Theoretic Neural Optimizer

AE4803 Robotic Systems and Autonomy, Georgia Tech (guest lecture)	10/2022
Rough Path Interest Group, Alan Turing Institute	12/2021
Machine Learning PhD Seminar, Georgia Tech (contributed talk)	10/2021
NeurIPS Workshop on Optimization for Machine Learning (spotlight talk)	12/2020

ACADEMIC SERVICES

Co-organizer:	NeurIPS 2025 Workshop on Frontiers in Probabilistic Inference: Sampling Meets Learning ICML 2024 Workshop on Structured Probabilistic Inference & Generative Modeling ICML 2023 Workshop on New Frontiers in Learning, Control, and Dynamical Systems
Area Chair:	NeurIPS 2023 Workshop on AI for Scientific Discovery: From Theory to Practice
Reviewer:	ICLR (2024 – 2025), ICML (2023 – 2025), NeurIPS (2023 – 2025), L4DC (2023 – 2024), IJCAI (2024), AISTATS (2025)